

StudyTracker - Pomodoro Study App

Architecture Overview

StudyTracker is an Android application implementing a Pomodoro timer system with persistent data storage, theme switching, and visual analytics. The app follows MVVM architecture with Room database, LiveData, and Foreground Services.

Tech Stack: Java, Room 2.6.0, Lifecycle 2.7.0, Material 1.11.0, AndroidX AppCompat 1.6.1

Themes: Focus (dark, low distraction) and Review (light, high contrast)

File Documentation

1. BaseActivity.java

Package: com.dandroids.studytracker.activities

Purpose: Abstract base class for theme switching and notification permissions.

Key Methods:

- toggleTheme() - Switches between Focus/Review themes, saves preference, recreates activity
- requestNotificationPermission() - Requests POST_NOTIFICATIONS on Android 13+

SharedPreferences: StudyTrackerPrefs with key is_focus_theme (default: true)

2. PomodoroManager.java

Package: com.dandroids.studytracker.manager

Purpose: Manages Pomodoro cycle with automatic state persistence.

Cycle Pattern:

Focus (25min) → Short Break (5min) → Focus → Short Break → Focus → Short Break → Focus → Long Break (30min) → reset

Constants:

Constant	Value
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FOCUS_DURATION	25 min
SHORT_BREAK_DURATION	5 min
LONG_BREAK_DURATION	30 min
CYCLES_BEFORE_LONG	4

Public Methods:

- advance() - Moves to next session type, saves state
- getCurrentDuration() - Returns duration in ms
- getCurrentLabel() - Returns human-readable label
- reset() - Resets to cycle 1, Focus type
- getCurrentType() - Returns SessionType enum
- getCurrentCycle() - Returns current cycle (1-4)

Persistence: Saves to PomodoroPrefs SharedPreferences

3. StudyViewModel.java

Package: com.dandroids.studytracker.viewmodel

Purpose: AndroidViewModel providing lifecycle-aware data to UI components.

LiveData Exposed:

- allSessions: LiveData<List<Session>>
- allSubjects: LiveData<List<Subject>>

Key Methods:

- insertSession(Session) / updateSession(Session)
- insertSubject(Subject) / deleteSubject(Subject)
- deleteAllSessions()
- getSessionsBySubject(long subjectId) - Returns LiveData filtered by subject
- getAllSessionsSync() - Synchronous query for exports

Threading: Single-threaded ExecutorService for all DB operations

4. AppDatabase.java

Package: com.dandroids.studytracker.db

Purpose: Room Database singleton. Version 1.

Entities: Session, Subject

DAOs: SessionDao, SubjectDao

Singleton Pattern:

```
AppDatabase db = AppDatabase.getInstance(context);
```

5. Session.java (Model)

Package: com.dandroids.studytracker.model

Table Structure:

Column	Type	Description
id	long	PRIMARY KEY, auto-generate
subjectId	long	FOREIGN KEY → Subject(id), ON DELETE CASCADE
startTime	long	Session start timestamp (ms)
endTime	long	Session end timestamp (ms)
durationMinutes	int	Total duration in minutes
completed	boolean	Whether session finished

Constructor:

```
Session(long subjectId, long startTime)
```

Helper:

```
long getDurationMillis() // Returns endTime - startTime
```

6. SessionDao.java

Package: com.dandroids.studytracker.db

Queries:

Method	Return Type	Description
insert(Session)	long	Returns row ID

update(Session)	void	Updates session
getAllSessions()	LiveData<List>	All sessions, newest first
getSessionsBetween(long, long)	LiveData<List>	Date range filter
getSessionsBySubject(long)	LiveData<List>	Filter by subject
getAllSessionsSync()	List	Synchronous version
deleteAll()	void	Clears all sessions

7. TimerService.java

Package: com.dandroids.studytracker.service

Purpose: Foreground service for Pomodoro timer. Supports binding for UI updates.

Constants:

Constant	Value
CHANNEL_ID	"study_timer_channel"
ACTION_PAUSE	"ACTION_PAUSE"
ACTION_STOP	"ACTION_STOP"
NOTIFICATION_ID	1

Key Methods:

- startTimer(long durationMillis) - Starts countdown
- pauseTimer() / resumeTimer() - Control timer
- stopTimer() - Stops completely
- setCallback(TimerCallback) - Register for tick/finish events
- isRunning() / getRemainingMillis() - Status methods

TimerCallback Interface:

```
void onTick(long millisRemaining);
void onFinish();
```

Features:

- Foreground notification with Pause/Stop actions
- Vibration pattern on completion (3 pulses)
- Binder pattern for Activity binding (TimerBinder class)
- Sticky service

Notification Action: Clicking notification opens SessionActivity

8. BarChartView.java

Package: com.dandroids.studytracker.views

Purpose: Custom view displaying weekly study minutes as a bar chart (Monday-Sunday).

Public Method:

```
void setSessions(List<Session> sessions)
```

Filters sessions to current week and aggregates durationMinutes per day.

Visual:

- Mon-Fri: Blue bars (#5C6BC0)
- Sat-Sun: Lighter blue (#9FA8DA)
- Minute labels displayed above bars
- Dynamic scaling based on max value

9. TimerProgressView.java

Package: com.dandroids.studytracker.views

Purpose: Custom circular progress indicator with timer text display.

Public Methods:

- `setProgress(long remainingMillis, long totalMillis)` - Updates progress arc and center text (MM:SS)
- `setLabelText(String label)` - Sets text below timer

Color Coding:

Progress	Color
> 50%	Blue (#5C6BC0)
20% - 50%	Orange (#FF9800)
< 20%	Red (#F44336)

Display:

- Large center: Time remaining (MM:SS)

- Small below: Session label (e.g., "Focus")
- Circular progress ring

10. themes.xml

Path: res/values/themes.xml

Theme	Parent	Background	Accent
Theme.StudyTracker (Focus)	DarkActionBar	#121428	#5C6BC0
Theme.StudyTracker.Review	Light.DarkActionBar	#FAFAFA	#3F51B5

Switching: BaseActivity.toggleTheme()

11. colors.xml

Path: res/values/colors.xml

Focus Theme:

- primary: #1B1F3B, background: #121428, accent: #5C6BC0, surface: #1E2240, text: #E8EAF6

Review Theme:

- primary: #FFFFFF, background: #FAFAFA, accent: #3F51B5, surface: #FFFFFF, text: #212121

Shared:

- timer_running: #4CAF50, timer_paused: #FF9800, chart_bar: #5C6BC0, chart_bar_alt: #9FA8DA

12. AndroidManifest.xml

Path: /app/src/main/AndroidManifest.xml

Permissions:

```
<uses-permission android:name="android.permission.VIBRATE" />
<uses-permission android:name="android.permission.FOREGROUND_SERVICE" />
<uses-permission android:name="android.permission.FOREGROUND_SERVICE_DATA_SYNC" />
```

```
<uses-permission android:name="android.permission.POST_NOTIFICATIONS" />
```

Activities:

Activity	Exported	Launch Mode
DashboardActivity	true (LAUNCHER)	standard
SessionActivity	false	singleTop
SessionDetailActivity	false	standard
SettingsActivity	false	standard

Service:

```
<service android:name=".service.TimerService"
    android:foregroundServiceType="dataSync"
    android:exported="false" />
```

FileProvider: Configured for file sharing with authority \${applicationId}.fileprovider

Dependencies

```
dependencies {
    implementation "androidx.room:room-runtime:2.6.0"
    annotationProcessor "androidx.room:room-compiler:2.6.0"
    implementation "androidx.lifecycle:lifecycle-viewmodel:2.7.0"
    implementation "androidx.lifecycle:lifecycle-livedata:2.7.0"
    implementation "com.google.android.material:material:1.11.0"
    implementation "androidx.appcompat:appcompat:1.6.1"
    implementation "androidx.core:core-ktx:1.12.0"
}
```

Quick Start Guide

Initialize Database:

```
AppDatabase db = AppDatabase.getInstance(context);
```

Start Timer Session:

```
Intent intent = new Intent(context, TimerService.class);
intent.putExtra("DURATION_MILLIS", PomodoroManager.FOCUS_DURATION);
ContextCompat.startForegroundService(context, intent);
```

Bind to Timer Service:

```
bindService(intent, serviceConnection, Context.BIND_AUTO_CREATE);
```

Save Completed Session:

```
Session session = new Session(subjectId, startTime);
session.endTime = System.currentTimeMillis();
session.durationMinutes = (int)((session.endTime - session.startTime) / 60000);
session.completed = true;
viewModel.insertSession(session);
```

Switch Theme:

```
toggleTheme(); // Activity recreates automatically
```

Data Flow Diagram

```
SessionActivity → TimerService → PomodoroManager
      ↓
    StudyViewModel
      ↓
    Room Database
      ↓
DashboardActivity ← BarChartView
```

Version History

Version	Date	Description
1.0	Current	Pomodoro timer, dual themes, Room persistence, weekly charts